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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2026

Time: 03 Hours

Code: 100202 ***Subject: Engineering Mathematics II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[2×7=14]**

(a) The harmonic conjugate of the function $u(x, y) = x^2 - y^2$ is

(i) $2xy$ (iii) $x^2 + y^2$
(ii) $-2xy$ (iv) xy

(b) If a function is analytic inside and on a simple closed contour C, then $\oint_C f(z)\,dz$

(i) $2\pi i$ (iii) ∞
(ii) 0 (iv) Depends on the contour

(c) The residue of the function $f(z) = e^z/z$ at $z = 0$ is

(i) 0 (iii) 1
(ii) e (iv) $2\pi i$

(d) The auxiliary equation of $y'' - 5y' + 6y = 0$ is

(i) $m^2 + 5m + 6 = 0$ (iii) $m^2 - 6m + 5 = 0$
(ii) $m^2 - 5m + 6 = 0$ (iv) $m^2 + 6m - 5 = 0$

(e) The standard form of a Cauchy–Euler differential equation is

(i) $y'' + py' + qy = r(x)$ (iii) $y'' + x^2y = 0$
(ii) $x^2y'' + axy' + by = 0$ (iv) $y' + xy = 0$

(f) The series $\Sigma_{n=1}^{\infty} 1/n^p$ converges if

(i) $p < 1$ (iii) $p > 1$
(ii) $p = 1$ (iv) $p \le 1$

(g) Cauchy's Condensation Test can be applied to

(i) Any alternating series (iii) Positive, decreasing series
(ii) Any finite series (iv) Oscillating series

(h) The Laplace Transform of 2^t is

(i) $\ln 2$ (iii) $1/(s - \ln 2)$
(ii) $1/(s - 2)$ (iv) $1/s$

(i) The inverse Laplace Transform of $1/(s^2 + 9)$ is

(i) $\sin 3t$ (iii) $3 \sin 3t$
(ii) $(1/3) \sin 3t$ (iv) $\cos 3t$

(j) The half-range cosine series is used when the function is extended as a

(i) Periodic function (iii) Even function
(ii) Odd function (iv) Constant function

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Q.2 (a) Prove that the function $f(z) = z^3$ is analytic. **[7]**

(b) Show that the function $u(x, y) = e^x(x \cos y - y \sin y)$ is harmonic. Hence, find its harmonic conjugate $v(x, y)$ and obtain the corresponding analytic function $f(z)$. **[7]**

Q.3 (a) Evaluate the contour integral $\oint_C (2z+1)/[(z-1)(z-2)]\, dz$, where C is the circle $|z - 1| = 2$, using the Cauchy Integral Formula. **[7]**

(b) Find the Laurent series expansion of the function $f(z) = 1/[z(z-1)]$ valid in the region $0 < |z| < 1$. Hence, determine the nature of the singularity at $z = 0$. **[7]**

Q.4 (a) Evaluate the definite integral $\int_0^{2\pi} d\theta/(5 + 4 \cos \theta)$, using the method of residues. **[7]**

(b) Solve the differential equation $y'' - 4y' + 4y = xe^{2x}$. **[7]**

Q.5 (a) Solve the Cauchy–Euler equation $x^2y'' + 5xy' + 4y = x^3$. **[7]**

(b) Using the Method of Variation of Parameters, solve the differential equation $y'' + y = \sec x$. **[7]**

Q.6 (a) Test the convergence of $\Sigma_{n=1}^{\infty} 2^n/n!$ using D'Alembert's Ratio Test. **[7]**

(b) Apply Raabe's Test to determine the nature of $\Sigma_{n=1}^{\infty} (2n)!/[4^n(n!)^2]$. **[7]**

Q.7 (a) Find the inverse Laplace Transform of $(s) = (2s+5)/[(s-1)(s^2+4)]$. **[7]**

(b) Solve the initial value problem using the Laplace Transform $y'' + 5y' + 6y = e^{-2t}$, where $y(0) = 1$, $y'(0) = 1$. **[7]**

Q.8 (a) Find the Fourier series of the function $f(x) = x$, where $-\pi < x < \pi$. Hence, deduce the value of $1 - 1/3 + 1/5 - 1/7 + - \cdots\cdots$ **[7]**

(b) Obtain the half-range sine series of $f(x) = x(\pi - x)$, where $0 < x < \pi$. **[7]**

Q.9 Write short note on any two of the following: **[7×2=14]**

(i) Cauchy–Riemann Equation

(ii) Laurent's Series

(iii) Convolution Theorem

(iv) Parseval's theorem

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